SHEAVES ARE EQUIVALENT TO ABSOLUTE GLUEING DATA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction: the values, the restriction maps, the presheaf property, the comparison isomorphisms and the sheaf property are all written out. The closing sentence 'Further details omitted' concerns the routine verification of the quasi-inverse property and is preserved verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Absolute glueing data. Same chapter, the definition of absolute glueing data used in the statement. Supplies the data which is converted into a sheaf.
- The compatibility condition of absolute glueing data. Same chapter, the compatibility condition satisfied by the transition isomorphisms of absolute glueing data. Ensures both that the constructed object is a presheaf and that the comparisons are isomorphisms.

Lemma 0.1. *Let $\mathcal{C}$ be a site. The category $Sh(\mathcal{C})$ is equivalent to the category of absolute glueing data via the functor that associates to $\mathcal{F}$ on $\mathcal{C}$ the canonical absolute glueing data.*

Proof. Given an absolute glueing data $(\mathcal{F}_U, c_f)$, we construct a sheaf $\mathcal{F}$ on $\mathcal{C}$ by setting $\mathcal{F}(U) = \mathcal{F}_U(U)$, where restriction along $f : V \to U$ given by the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}_U(U) & \longrightarrow & \mathcal{F}_U(V) & \xrightarrow{c_f} & \mathcal{F}_V(V) \\ \| & & & & \| \\ \mathcal{F}(U) & & \longrightarrow & & \mathcal{F}(V) \end{array}$$

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

The compatibility condition $c_g \circ j_g^{-1} c_f = c_{f \circ g}$ ensures that $\mathcal{F}$ is a presheaf, and also ensures that the maps $c_f : \mathcal{F}_U(V) \to \mathcal{F}(V)$ define an isomorphism $\mathcal{F}_U \to \mathcal{F}|_{\mathcal{C}/U}$ for each U. Since each $\mathcal{F}_U$ is a sheaf, this implies that $\mathcal{F}$ is a sheaf as well. The functor $(\mathcal{F}_U, c_f) \mapsto \mathcal{F}$ just constructed is quasi-inverse to the functor which takes a sheaf on $\mathcal{C}$ to its canonical glueing data. Further details omitted. $\square$